When the Exchange Rate Moves

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Dean’s Professor

Module 3: How border costs reach the firm

Panama · Course photograph

September 8
Distance and logistics

Today
Exchange rates and currency mismatch

September 14
Legal jurisdictions and tax

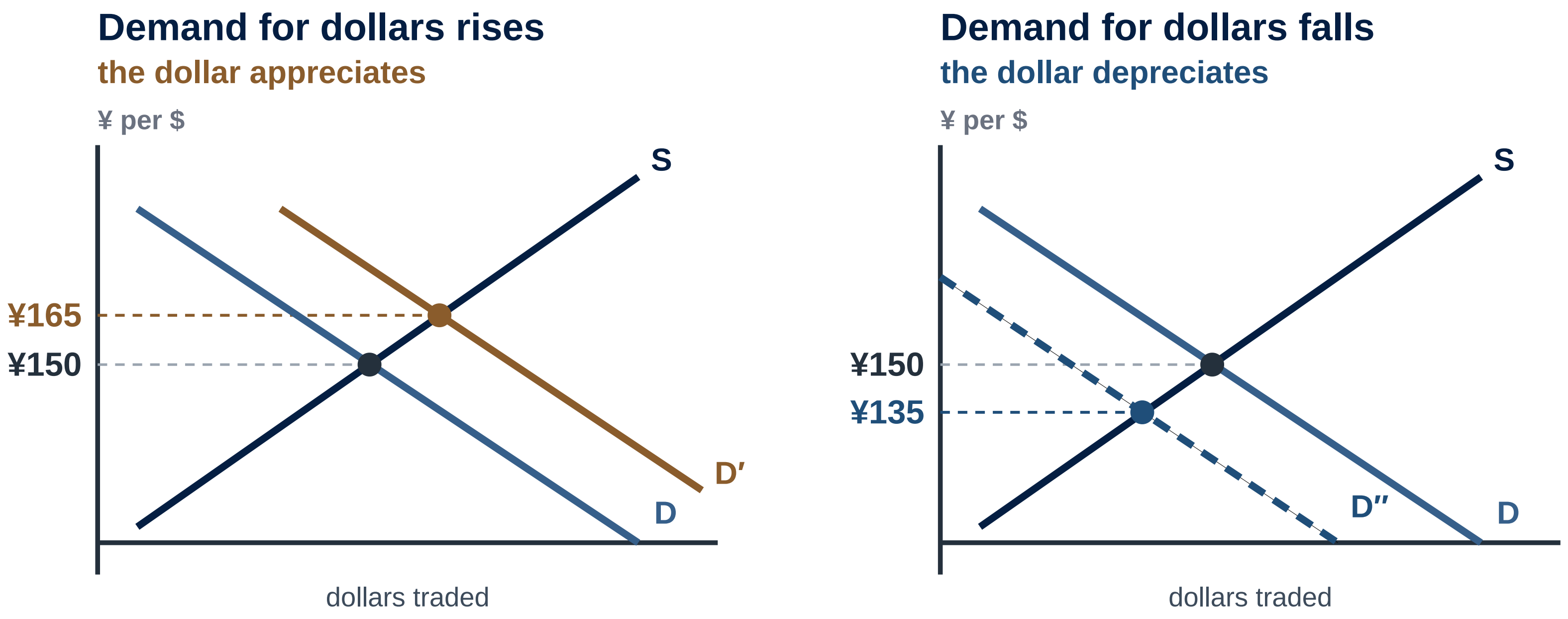
> The border can lower a shipping cost, change the value of a debt, or determine where profit is taxed.

A currency mismatch could erase the gains from cheaper distance

A Thai factory in 1996: baht wages, dollar loans, a fixed exchange rate.

> Containerization reduced the cost of distance. A currency mismatch could erase the gain.

What sets a nominal exchange rate



Demand · Japanese buyers of US goods and assets need dollars.

Supply · Americans need yen for Japanese goods and assets.

Demand rises when US rates rise, US goods sell better, or investors move into dollar assets.

> More yen per dollar means the dollar has appreciated and the yen has depreciated.

The IMF lends when a country runs out of foreign currency

- Temporary financing goes to member governments facing external-payment problems.
- The government and IMF negotiate policy conditions intended to restore reserves and access to financing.
- The IMF does not rescue individual firms or repay their private loans.

> The dispute is not whether Thailand needed dollars. It is what policies should accompany them.

IMF headquarters · Ajay Suresh / Wikimedia Commons · CC BY 2.0

A Brief Writing Exercise

YOUR IB VOICE

QUESTION

Thai firms borrowed dollars because dollar loans were cheaper than baht loans, and few of them hedged. Was that a mistake or a reasonable bet that went wrong? Who bears most responsibility for what followed: the firms, the Bank of Thailand, foreign lenders, hedge funds, or the IMF? Defend your answer with evidence from Blustein.

10 minutes · Individual · Notes permitted

Commit to an answer before we reconstruct the sequence.

What evidence would show that dollar borrowing was a mistake rather than a reasonable bet?

> Which actor had the best information and the greatest ability to prevent the crisis?

The Bet

A fixed rate made dollar debt look safe

25

baht per dollar, held for more than a decade

Every morning the Bank of Thailand set the rate and bought or sold unlimited dollars at it.

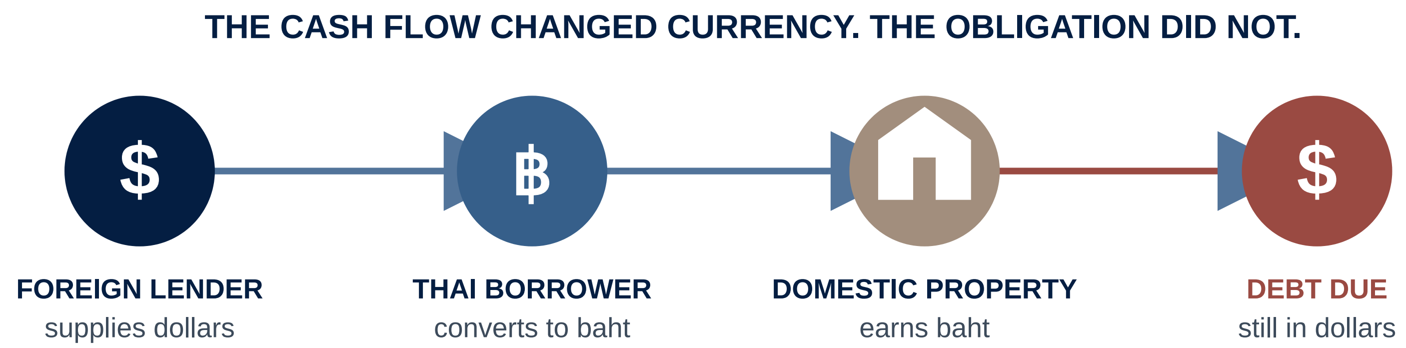
Japanese carmakers and Thai exporters planned around it.

> A fixed rate removes the exchange-rate calculation from every decision, until it breaks.

Dollar borrowing created a currency mismatch

$56 billion was borrowed through the Bangkok International Banking Facilities by 1997, three times the 1994 level.

The facility, launched in 1992 with IMF support, made foreign-currency borrowing cheaper.



> Borrowers and lenders treated the peg as protection.

Cheap foreign money flowed into property and stocks

Credit to Thailand’s private sector: **$89 billion in 1992, $204 billion in 1996.**

Bangkok office space added in 1996: **600,000 square metres, half empty.**

Official bad-loan rate by May 1997: **12 percent.**

> The assets earned baht. The debts were owed in dollars.

Sathorn Unique Tower · Chainwit. / Wikimedia Commons · CC BY-SA 4.0

As the dollar rose, Thai export growth stalled

Pegged to a rising dollar from 1995, Thai clothes, shoes, and electronics became more expensive on world markets.

Export growth in 1996: **flat**. Current account deficit: **8 percent of GDP**.

> A fixed rate imports the movements of the currency it is fixed to.

The Defense

The warnings came before the reserves ran out

IMF warnings: a January letter, visits in March and May, and advice to adjust the rate and widen the band.

The refusal. Chaiyawat Wibulswasdi, deputy governor and the chapter's Winnie the Pooh, argued that Mexico had tried a controlled devaluation and failed. If 25 will not hold, why would 28, or 30?

> The Bank of Thailand held the peg while its room to defend it disappeared.

The reserve headline concealed the cost of defense

EARLY 1997

$38B

published reserves

the visible headline

LATE MAY

$23B

committed forward

an off-balance-sheet claim

JUNE

≈$2B

usable reserves

after the defense continued

These are checkpoints, not parts of one accounting total.

Thai ministers who asked for the number were refused by their own central bank. Fischer was given it in late July, on condition that he tell no one else at the Fund.

> The reported reserve figure did not show how much cash remained available.

Capital controls stopped the shorts, not the dollar borrowers

Hedge fund

Borrow baht → sell baht → borrowing blocked

Attack interrupted

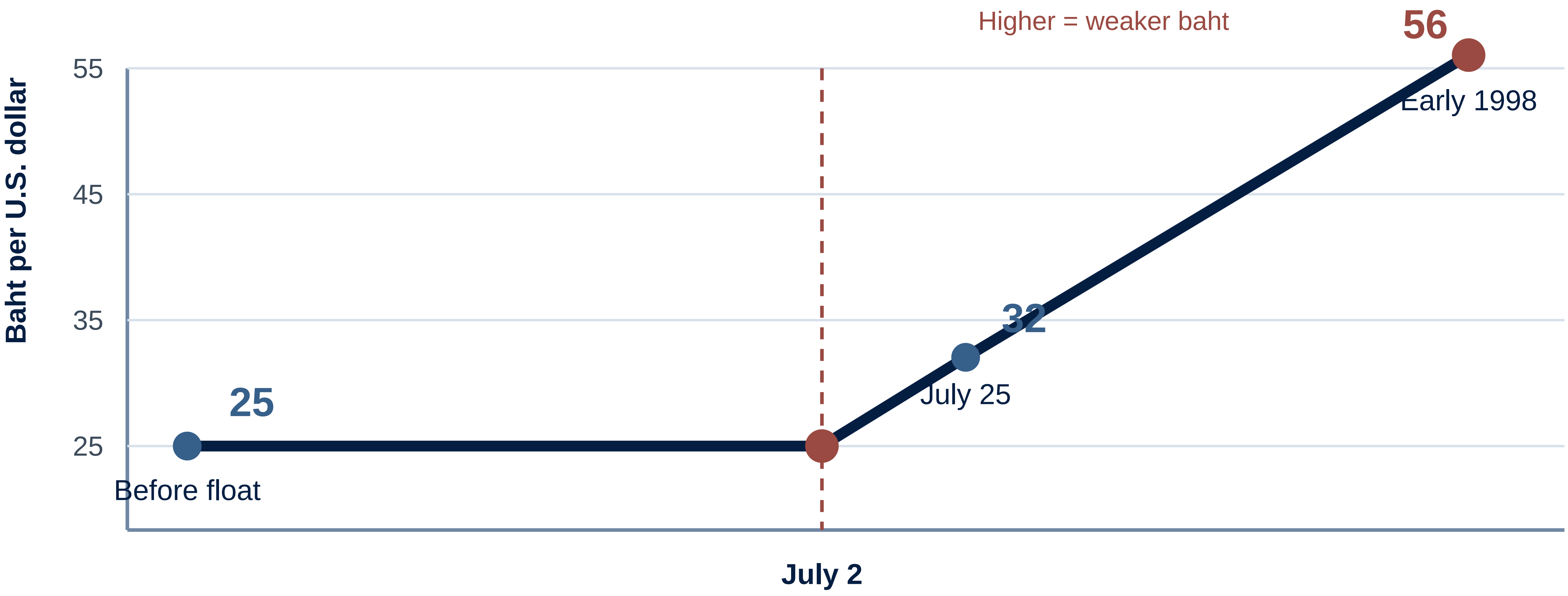
Thai borrower

Dollar loan due → obtain dollars → sell baht

Pressure continued

> Beating the speculators did not stop the run.

The baht's collapse more than doubled dollar debt



Selected milestones from the case. The connecting line shows the direction of change, not daily observations.

> A $1 million loan went from 25 million to 56 million baht. Revenue in baht did not move.

Your Industry Teams

Industry teams begin today

Teams are assigned by the list distributed now. You stay with your team through December.

Today’s task: a Thai firm in 1997, then one exposure in your own industry.

5 minutes · Find your team and sit together

Rerun the year for one firm

Line	Per year
Revenue, sold in Thailand	700 million baht
Revenue, exported	$12 million
Imported inputs	$16 million
Local wages and costs	250 million baht
Dollar loan due this year	$20 million

> Compute operating profit before debt repayment and the baht cost of repaying principal at 25, 32, and 56 baht per dollar. Then decide: what could this firm have hedged, and what could it not?

12 minutes · Teams · One sheet per team

Which of this firm’s losses were exchange-rate risk, and which were something else?

> Each team reports its profit at 56, one thing the firm should have hedged, and one exposure in its own industry.

Aftermath

The rescue demanded austerity from a collapsing economy

−10%

Thai GDP in 1998; unemployment tripled, urban wages fell 8 percent

Japan and China were asked for bilateral loans first, and refused.

$17.2 billion package, none of it from the United States.

The IMF required tax rises and spending cuts of 3 percent of GDP.

> The current account swung from 8 percent deficit to 12.7 percent surplus in a year, by collapse.

The same shock, different firms

Dollar revenue and baht costs: gained. Baht revenue and dollar debt: failed.

Hedging covers a known payment on a known date, not a creditor who will not renew.

> Exchange-rate risk is a mismatch between where a firm earns and where it owes.

Who was responsible?

> Monday: one firm, many legal jurisdictions. How Apple’s tax structure worked, and the $17 billion Ireland was ordered to collect.